



INSTITUTO POLITÉCNICO NACIONAL
UNIDAD PROFESIONAL INTERDISCIPLINARIA DE
INGENIERÍA Y TECNOLOGÍAS AVANZADAS



CARRERA: INGENIERÍA TELEMÁTICA

UNIDAD DE APRENDIZAJE: MULTIMEDIA

PRACTICA: 8-INTEGRACION SIT CON EL TTS (PARTE 1)

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DOCENTE: NOE SIERRA ROMERO

FECHA: 24/02/26

GRUPO: 2TV5



-CODIGO:

```
<!DOCTYPE html>
<html lang="es">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Texto a Voz Simplificado</title>
</style>
body {
  font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
  max-width: 500px;
  margin: 50px auto;
  padding: 20px;
  background: #f4f7f6;
  border-radius: 12px;
  box-shadow: 0 4px 15px rgba(0,0,0,0.1);
}
h2 { text-align: center; color: #2c3e50; }

textarea {
  width: 100%;
  height: 120px;
  padding: 12px;
  border: 2px solid #ddd;
  border-radius: 8px;
  resize: none;
  box-sizing: border-box;
  font-size: 16px;
}

.controls {
  margin-top: 20px;
  display: flex;
  flex-direction: column;
  gap: 15px;
}

.control-group {
  display: flex;
  flex-direction: column;
  gap: 5px;
}

label { font-weight: bold; color: #555; font-size: 14px; }

select, input[type="range"] {
  width: 100%;
  cursor: pointer;
}

#btn-speak {
  width: 100%;
  margin-top: 25px;
  padding: 15px;
  background: #4CAF50;
  color: white;
  border: none;
  border-radius: 8px;
  font-size: 18px;
  font-weight: bold;
  cursor: pointer;
  transition: background 0.3s;
}

#btn-speak:hover { background: #45a049; }

#status {
  margin-top: 15px;
  text-align: center;
}
```

```

    font-size: 14px;
    color: #888;
  }
</style>
</head>
<body>

< h2 > ÷ Lector de Texto</h2>

<textarea id="input-area" placeholder="Escribe tu mensaje aquí..."></textarea>

<div class="controls">
  <div class="control-group">
    <label>Seleccionar Voz:</label>
    <select id="voice-select"></select>
  </div>

  <div class="control-group">
    <label>Velocidad: <span id="rate-val">1</span>x</label>
    <input type="range" id="rate" min="0.5" max="2" step="0.1" value="1">
  </div>

  <div class="control-group">
    <label>Tono (Pitch): <span id="pitch-val">1</span></label>
    <input type="range" id="pitch" min="0" max="2" step="0.1" value="1">
  </div>
</div>

<button id="btn-speak" onclick="speak()">▶ PLAY</button>

<div id="status">Listo para reproducir</div>

<script>
const synth = window.speechSynthesis;
const voiceSelect = document.getElementById('voice-select');
const rateInput = document.getElementById('rate');
const pitchInput = document.getElementById('pitch');
const status = document.getElementById('status');
let voices = [];

function loadVoices() {
  voices = synth.getVoices();
  voiceSelect.innerHTML = "";
  voices.forEach((voice, i) => {
    const option = document.createElement('option');
    option.value = i;
    option.textContent = `${voice.name} (${voice.lang})`;
    voiceSelect.appendChild(option);
  });
}

if (speechSynthesis.onvoiceschanged !== undefined) {
  speechSynthesis.onvoiceschanged = loadVoices;
}
loadVoices();

// Actualizar visualmente los valores de los sliders
rateInput.oninput = () => document.getElementById('rate-val').innerText = rateInput.value;
pitchInput.oninput = () => document.getElementById('pitch-val').innerText = pitchInput.value;

function speak() {
  const text = document.getElementById('input-area').value;
  if (!text) return;

  // Cancelar cualquier audio que se esté reproduciendo actualmente
  synth.cancel();

  const utterance = new SpeechSynthesisUtterance(text);

  // Aplicar configuraciones elegidas

```

```

utterance.voice = voices[voiceSelect.value];
utterance.rate = rateInput.value;
utterance.pitch = pitchInput.value;

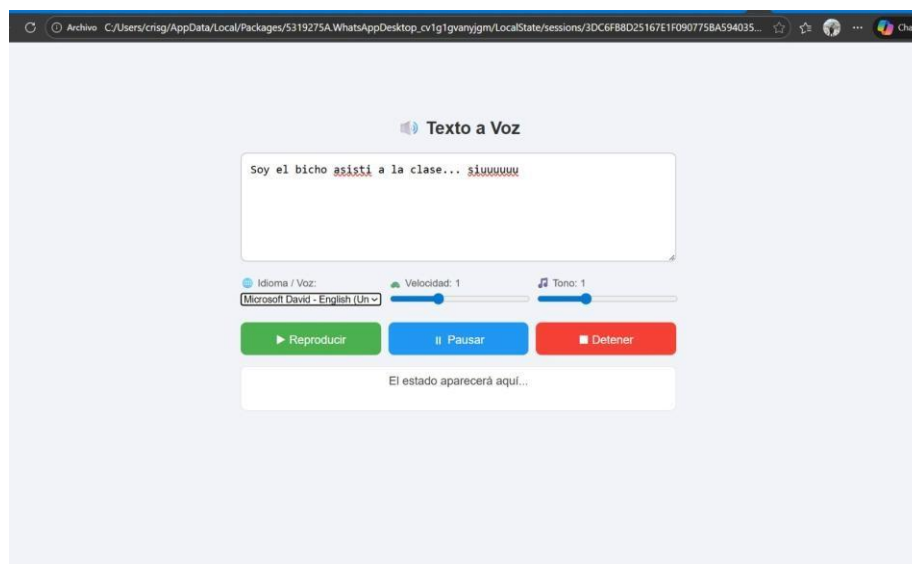
utterance.onstart = () => status.innerText = "Reproduciendo...";
utterance.onend = () => status.innerText = "Finalizado.";

synth.speak(utterance);
}
</script>

</body>
</html>

```

Conversor de texto principal:



Cambio realizados:



